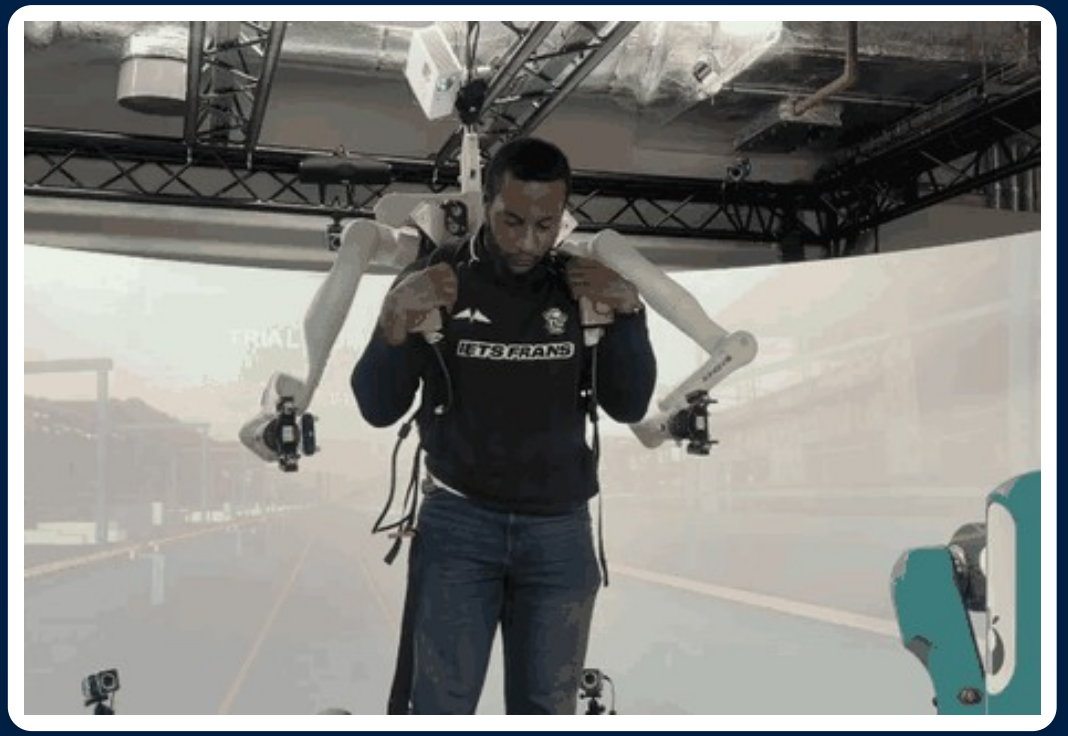
Imperial College London

# World-Stable Supernumerary Effectors for Human Augmentation Under Locomotion-Scale Base Motion

Keeping a wearable robot's hands still while the wearer walks

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Watch the hands, not the elbows.

# Every step moves your extra hands



**63–68 mm**

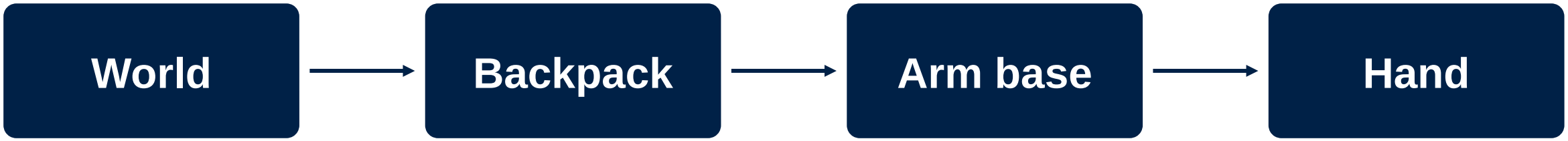
hand motion, arm locked (RMS)

**~1.8 Hz**

step rate

**Goal: hands fixed in the room, not the body**

# Measure the backpack, cancel it through the arm




Vicon, live

fixed offset

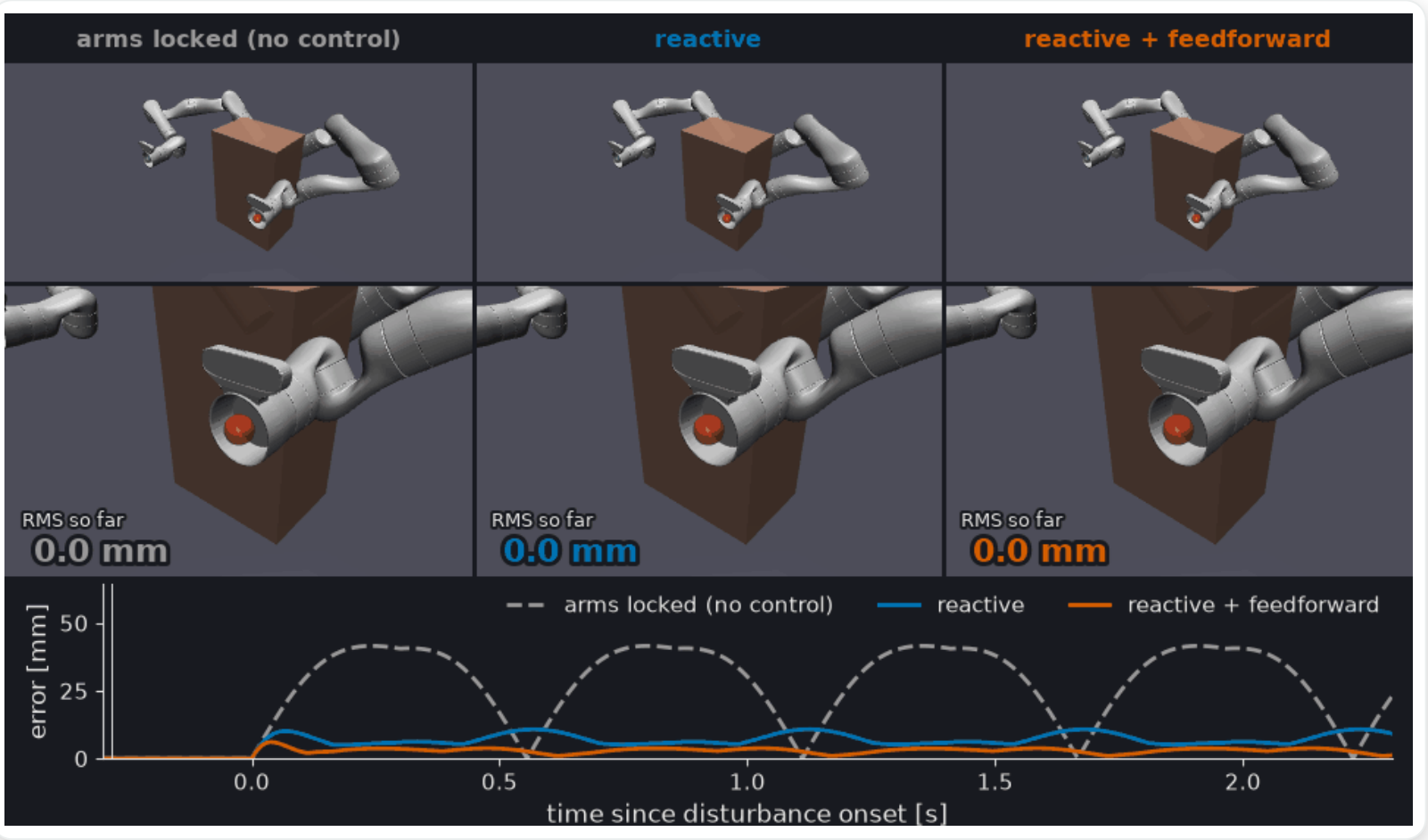
forward kinematics

$$\dot{q} = J^{+}_{\lambda}\,(K_p \cdot e + K_d \cdot \dot{e}) \;+\; (I - J^{+}J)\cdot \dot{q}_0$$

room-frame error → joint speeds

▲ velocity term

# Simulation first: my controller in a physics engine




**Hand error, RMS over 8 s**

**31.0 mm**

arms locked

**6.7 mm**

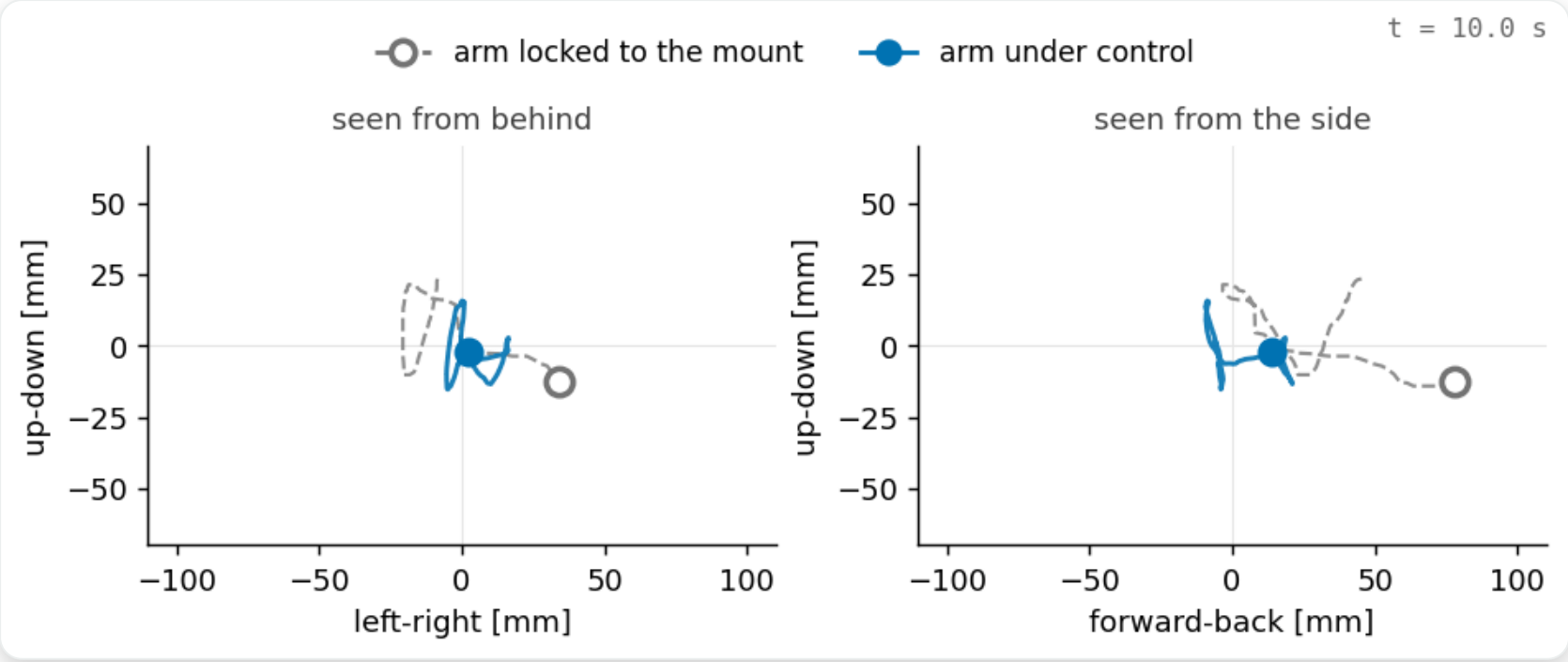
reactive

**2.6 mm**

reactive + feedforward

# On real wearers: about 2/3 of walking motion removed

Trials: a 400 Hz controller on the lab machine. Analysis: mine.




**71%**

0.5 m/s

**63%**

1.0 m/s

**56%**

1.5 m/s

Mean of 6 wearers

# Turning up the gain barely helped

One-wearer pilot: $K_d$ 0.8 → 2.6. Response near 1 Hz fell:

**38%**

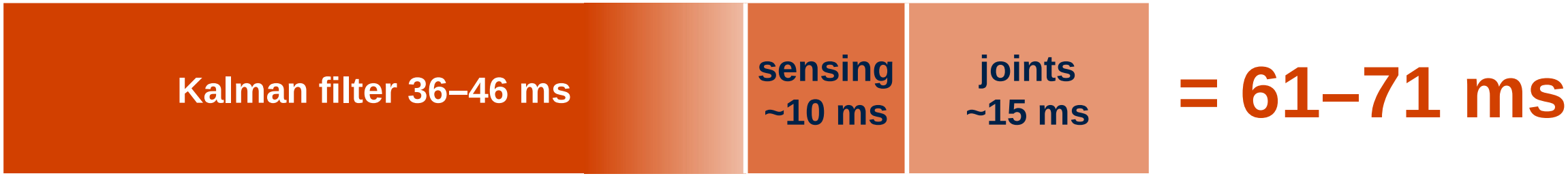
predicted

**9%**

measured

**The velocity term was barely acting. Why?**

# Best explanation: the velocity arrives 61–71 ms late




## Uncancelled: $2 \cdot \sin(\omega d/2) \approx 2/3$ at 1.8 Hz

Nothing fitted: beats no-velocity model, 6/6 wearers

# Impact, and what transfers

**2/3**

walking motion removed on hardware

**Next:** faster estimator

- Frames, calibration → camera extrinsics
- Latency budget → sensor time-sync
- Ground-truth the metric

# Thank you. Questions?

**Results, simulation, figures**

github.com/fechachris4/msc_project

**C++ real-time controller**

github.com/fechachris4/HumanSL_MAIN

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